Performane Evaluation of Small Language Models on Movie Review Dataset

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Overview

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Introduction

- Small Language Models (SLMs) enable NLP deployment in resource-constrained environments with low-latency requirements
- Architectural constraints of SLMs (limited parameters, context windows) create bottlenecks for structured knowledge integration
- Critical challenge: Effective assimilation of domain-specific knowledge from external sources like CSV files
- Current methodologies present significant trade-offs between efficiency and performance
- This study evaluates three integration paradigms: In-context Learning, RAG, and Fine-tuning on movie dataset

Definition of SLM

However there is no standard definition of a Small Language Model. Therefore we define a SLM as a model with parameters lesser than 1B.

Introduction

- Three knowledge integration paradigms are systematically evaluated:
  - **In context Learning**
  - **Retrieval Augmented Generation (RAG)**
  - **Fine tuning**

Research Objective

The objective of this work is to evaluate how different integration strategies affect SLM's specialization in movie-domain knowledge while maintaining their core reasoning competencies.

Key Challenge

The fundamental context window limitation of SLMs presents a significant bottleneck for effective assimilation of structured, domain-specific knowledge from external sources.

Motivation: The Rise of Small Language Models

Why Small Language Models?

- **Resource Constraints**: Enable deployment in environments with limited computational resources
- **Low Latency Requirements**: Critical for real-time applications and edge devices
- **Cost Efficiency**: Significantly lower operational costs compared to LLMs
- **Data Privacy**: On-device processing reduces data transmission risks

Key Use Cases for SLMs

Edge Computing and Mobile Applications

- Mobile applications with on-device AI capabilities
- IoT devices and embedded systems requiring low-power inference
- Real-time processing applications with strict latency requirements
- Offline capabilities for environments with limited connectivity

Motivation

Key Use Cases for SLMs

Specialized Domains and Enterprise Solutions

- Domain-specific virtual assistants and chatbots
- Structured data querying and database interaction systems
- Repetitive task automation in controlled environments
- Custom enterprise solutions with proprietary knowledge bases

The Critical Challenge

- Architectural constraints (limited parameters, context windows) create bottlenecks
- Effective integration of structured, domain-specific knowledge remains challenging
- Need for specialized models without compromising general capabilities

Definitions

In Context Learning

In-context learning is the ability of a language model to infer a task or pattern and generate appropriate outputs based solely on examples or instructions provided in the input context, rather than through gradient-based training.

Retrieval Augmented Generation

An approach that combines a retriever (finds relevant text chunks (from documents, CSVs, databases, etc.) and a generator (uses those chunks as context to generate the answer).

Fine Tuning

Fine-tuning is the process of taking a pre-trained model (like T5, BERT, GPT-Neo, etc.) and training it further on a smaller, domain-specific dataset so it adapts to a new task or style.

Related Works

Knowledge Distillation

1. [West et al., 2022]: Symbolic knowledge distillation from LLMs to create compact commonsense models
2. [Fu et al., 2023]: Specialized reasoning capabilities in smaller models via teacher-student distillation
3. [Zhang et al., 2024]: Chain-of-thought reasoning transfer from large to small models

Efficient Model Architectures

1. [Zhang et al., 2024]: TinyLlama - extensive pretraining on trillion tokens for compact models
2. [Song et al., 2023]: LLM-Planner for grounded planning in embodied agents
3. [Belcak et al., 2025]: SLMs are sufficiently powerful for agentic systems and offer 10-30$\times$ better inference efficiency

Contribution

Research Gap

Most existing approaches focus on reasoning capabilities; our work specifically addresses structured data integration for domain-specific knowledge in SLMs.

Evaluation tests both questions directly answerable from the dataset and general questions, assessing specialized knowledge acquisition and general proficiency retention. Our study extends beyond conventional benchmarks by:

- Evaluating on **domain-specific structured data** (movie metadata)
- Comparing **three integration paradigms** under identical conditions
- Measuring both **specialization effectiveness** and **general capability preservation**
- Assessing **computational efficiency** alongside accuracy metrics

Dataset Preparation

Dataset Composition

The dataset was constructed by merging and cleaning two primary sources:

- **MovieLens Dataset**: Standard movie rating and metadata collection
- **IMDB Spoiler Dataset**: Additional movie information from Kaggle

Data Processing Steps

1. Intersection of both datasets resulted in **3,459 unique movies**
2. Comprehensive cleaning to remove all null values and redundant entries
3. Structured formatting for consistent model processing
4. Validation to ensure data integrity and completeness

The final dataset contains **21 features** spanning movie metadata, financial information, production details, and audience metrics - providing comprehensive coverage for movie-related queries.

Dataset Structure (Part 1/4)

Table: Structure of the Cleaned Movie Dataset

Column Name	**Description**
movie_id	Unique identifier for each movie
title	Title of the movie
cast	List of main cast members and their roles
crew	Information about the movie crew (e.g., director, writer)
overview	Short summary of the movie plot
budget	Production budget of the movie (in USD)

Dataset Structure (Part 2/4)

Table: Structure of the Cleaned Movie Dataset (Continued)

Column Name	Description
genres	List of movie genres
homepage	Official website of the movie
keywords	Important keywords or tags related to the movie
original_language	Original language of the movie
popularity	Popularity score of the movie
production_companies	Companies involved in producing the movie

Dataset Structure (Part 3/4)

Table: Structure of the Cleaned Movie Dataset (Continued)

Column Name	**Description**
production_countries	Countries where the movie was produced
release_date	Release date of the movie
revenue	Box office revenue (in USD)
runtime	Duration of the movie (in minutes)
spoken_languages	Languages spoken in the movie

Dataset Structure (Part 4/4)

Table: Structure of the Cleaned Movie Dataset (Continued)

Column Name	**Description**
status	Release status (e.g., Released, Post Production)
tagline	Marketing tagline of the movie
vote_average	Average user rating
vote_count	Number of user votes

Proposed Methodology: In-Context Learning

Core Concept

- No parameter updates or external retrieval mechanisms
- Tests fundamental ability to utilize presented context
- Simulates user-provided reference documents

Key Challenges

- Context window limitations (C_{max} constraint)
- Random sampling variability
- Computational inefficiency for large datasets

Proposed Methodology: Retrieval-Augmented Generation

Two-Phase Architecture

Indexing Phase

- Convert dataset to vector embeddings
- Build searchable knowledge base
- Store in vector database $\mathcal{V}$

Online Query Phase

- Encode user query $\mathbf{q} = E(Q)$
- Retrieve top-k relevant entries
- Augment prompt with context
- Generate final response

Technical Components

- **Encoder**: E for text-to-vector conversion
- **Retrieval**: Semantic search with cosine similarity
- **Fusion**: $S_{context} = \text{concatenate}(D_{ret})$
- **Generation**: $R = M(P)$ with augmented context

Proposed Methodology: Fine-Tuning Approaches

Full Fine Tuning: Full fine-tuning means updating all the parameters (weights and biases) of a pre-trained model on your custom dataset.

Fine Tuning with LORA: LoRA is a parameter-efficient fine-tuning (PEFT) method that keeps the original model frozen and only trains a few small adapter matrices inserted into the model's layers.

Fine Tuning with QLORA: It combines quantization (compressing model weights) with LoRA adapters for ultra-efficient fine-tuning on consumer GPUs.

Training Data Generation Pipeline

We have generated a set of 50 Questions which is based on the csv which was being preprocessed. Sample Questions:

Movie Plot	**Correct Answer**
A team of superheroes assembling to stop an alien invasion	The Avengers
A young wizard who discovers his magical heritage	Harry Potter and the Sorcerer's Stone

QA Generation

- **Source**: Raw movie dataset $\mathcal{D}$ with 3,459 entries
- **Volume**: 50 Questions

Evaluation Metrices

- Accuracy: The percentage of times the model's predicted answer exactly matches the correct answer from eval_questions.csv

$$\text{Accuracy} = \frac{\text{Number of Exact Matches}}{\text{Total Number of Questions}} \times 100 \tag{1}$$

- Avg Latency: The average time (in seconds) the model takes to generate an answer for each question
- Avg Similarity: The average semantic similarity between the predicted and correct answers, measured using cosine similarity of sentence embeddings.

$$\text{Cosine Similarity}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}} \tag{2}$$

Experimental Setup

Installations: pandas, sentence-transformers, faiss-cpu, transformers
Library Imports: pandas, SentenceTransformer, util, pipeline, faiss, numpy, time
Systems Usage:
- Apple M2 Chip on Macbook Air
- Google Colab (T4 and CPU)
- Kaggle (2xT4 + CPU)

Results

Models Used:

- For Embeddings: `sentence-transformers/all-MiniLM-L6-v2` (33.4M parameters)
- For Answering: `deepset/minilm-uncased-squad2` (33.4M Parameters)

Question	Correct Answer	Predicted	Lat.	Sim.
Which movie has the plot about a team of superheroes assembling to stop an alien invasion?	The Avengers	Thriller	0.68	0.33
Which science fiction film shows humans fighting machines in a simulated reality?	The Matrix	The Matrix	0.37	1.00

Parameter	Result
Accuracy	12.00%
Average Latency (s)	0.03
Average Similarity	0.436

Results

Models Used:

- **Embedding:** `sentence-transformers/all-MiniLM-L6-v2` (33.4M parameters)
- **For Answering:** `deepset/roberta-base-squad2` (0.1B parameters)

Parameter	Result
Accuracy	20.00%
Average Latency (s)	0.04
Average Similarity	0.548

Question	Correct Answer	Predicted	Lat.	Sim.
Which movie has the plot about a team of superheroes assembling to stop an alien invasion?	The Avengers	Scary Movie	1.54	0.37
Which movie features Robert Downey Jr. as a genius billionaire inventor in a red-and-gold suit?	Iron Man	Iron Man 2	1.29	0.86

Results

Models Used:

- **Embedding:** `sentence-transformers/all-MiniLM-L6-v2` (33.4M parameters)
- **For Answering** `distilbert-base-uncased-distilled-squad` (65.2M parameters)

Parameter	**Result**
Accuracy	4.00%
Average Latency (s)	0.03
Average Similarity	0.350

Question	**Correct Answer**	**Predicted**	**Latency**	**Sim.**
Which science fiction film shows humans fighting machines in a simulated reality?	The Matrix	The Matrix	0.56	1.00
Which movie follows a group of friends trying to destroy a powerful ring in Middle-earth?	The Lord of the Rings: The Fellowship of the Ring	The Lord of the Rings	1.34	0.90

Results:

Models Used:

- **Embedding:** `sentence-transformers/all-MiniLM-L6-v2` (33.4M parameters)
- **QA Model:** `timpal0l/mdeberta-v3-base-squad2` (O.3B Parameters)

Parameter	Result
Accuracy	12.00%
Average Latency (s)	0.09
Average Similarity	0.440

Question	Correct Answer	Predicted Answer	Latency	Sim.
Which animated movie tells the story of toys that come to life when humans aren't looking?	Toy Story	Feast	4.26	0.17
Which animated movie follows a lion cub destined to be king of the Pride Lands?	The Lion King	The Lion King	4.09	1.00

Results

Models Used:

- **Embedding:** `sentence-transformers/all-MiniLM-L6-v2` (33.4M parameters)
- **QA Model:** `bert-large-uncased-whole-word-masking-finetuned-squad`

Parameter	Result
Accuracy	16.00%
Average Latency (s)	0.11
Average Similarity	0.495

Question	Correct Answer	Predicted	Latency	Sim.
Which movie follows a panda who becomes a kung fu warrior?	Kung Fu Panda	Kung Fu Panda	5.02	1.00
Which movie features a space ranger and cowboy doll becoming best friends?	Toy Story 2	America's Sweet-hearts	8.19	0.18

Results

Models Used:

- **Embedding:** `intfloat/e5-small-v2`
- **QA Model:** `deepset/minilm-uncased-squad2`

Parameter	Result
Accuracy	18.00%
Average Latency (s)	0.03
Average Similarity	0.852

Question	Correct Answer	Predicted	Latency	Similarity
Which Pixar film shows a robot left alone on Earth to clean up humanity's mess?	WALL·E	WALL·E	0.03	1.00
Which fantasy movie features a lion, a witch, and a magical wardrobe?	The Chronicles of Narnia: The Lion, the Witch and the Wardrobe	The Chronicles of Narnia: The Lion	0.03	0.97

Future Implementation

- Further Research on more Research papers on Small Language Models
- Considering models of different architectures like Mamba
- Implementation of Fine Tuning and Evaluation
- Implementing more Small Language models and Large Language Models for a better comparision and evaluation

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